

# SUPPLY CHAIN FULFILLMENT & LOGISTICS PERFORMANCE

Executive Report

## PROJECT OBJECTIVE

Evaluate delivery performance and identify operational factors contributing to late deliveries across regions, shipping modes, product categories, and customer segments.

## DATASET & TOOLS

DataCo Supply Chain Dataset  
SQL • Python • Pandas • Power BI  
Workflow: Extraction → Cleaning → EDA → Dashboard

## KEY PERFORMANCE INDICATORS

| Total Deliveries | On-Time Rate | Late Delivery Rate | Avg. Actual Days | Avg. Scheduled Days | Avg. Shipping Delay |
|------------------|--------------|--------------------|------------------|---------------------|---------------------|
| 180,519          | 17.9%        | 55.0%              | 3.5              | 2.9                 | 0.6 days            |

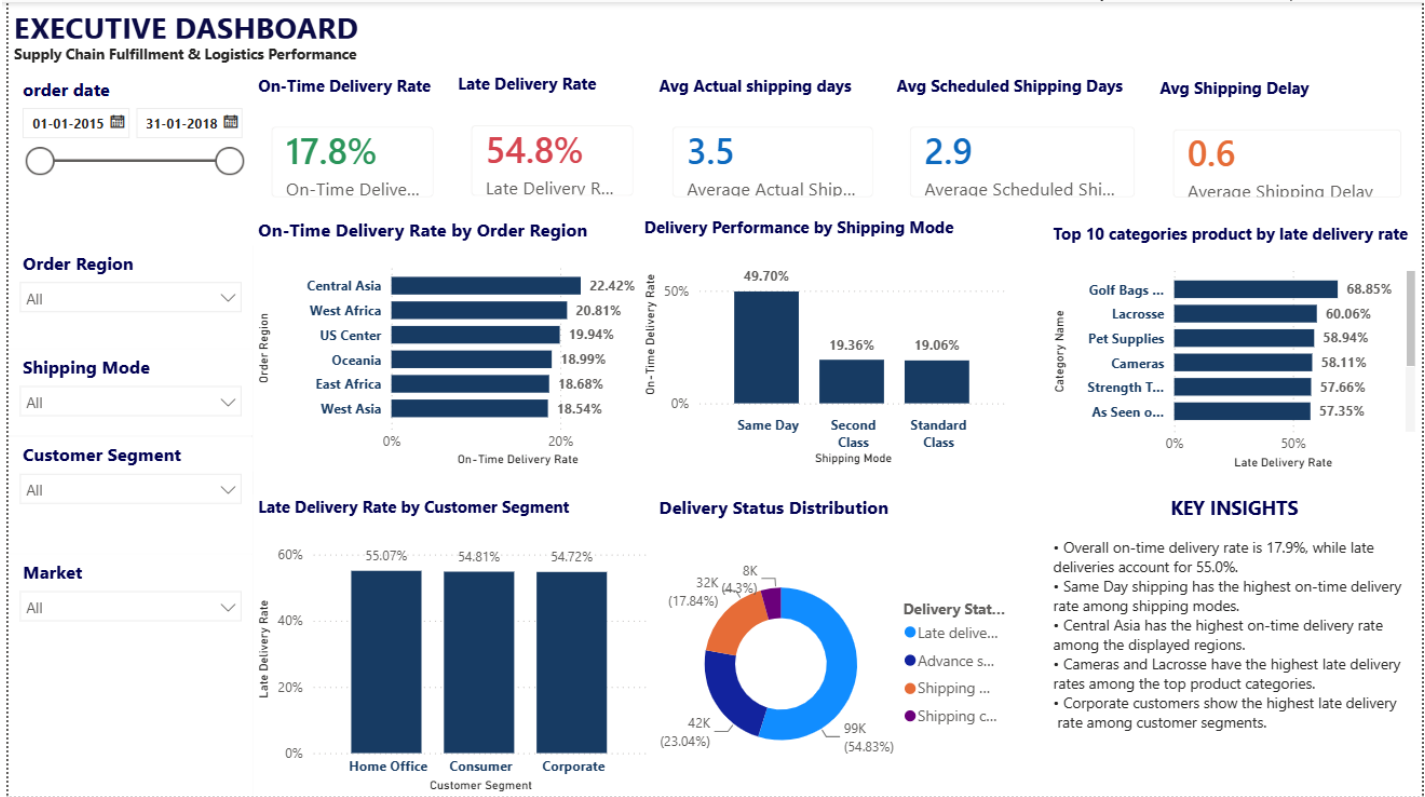
## KEY FINDINGS

- Delivery delays are the major operational issue, with late deliveries accounting for over 50% of all shipments.
- Central America and Western Europe have the highest order volumes, making them important markets for the company.
- Central Africa has the highest late-delivery rate at approximately 58%, followed by South Asia and East Africa.
- First Class and Second Class shipping show the poorest delivery performance, with late-delivery rates of approximately 95% and 77%, respectively.
- Same Day shipping performs comparatively better, with nearly half of shipments delivered on time.
- Standard Class is the most frequently used shipping mode, indicating the company's strong reliance on standard delivery.
- Actual shipping time does not consistently match scheduled shipping time, particularly for orders scheduled within 1–2 days.
- Late deliveries are widespread across major product categories, with Leather, Men's Footwear, and Women's Apparel having the highest absolute numbers of late deliveries.
- Late-delivery rates are consistently high across customer segments, suggesting that delays are not limited to a particular customer group.
- Only about one-third of orders are marked as Complete, while a substantial proportion remain in pending, payment, or processing stages.
- Delivery performance appears to be a broader operational challenge, rather than an issue limited to one region, shipping mode, or customer segment.

## BUSINESS RECOMMENDATIONS

- Improve delivery performance in high-risk regions by investigating the causes of delays in Central Africa, South Asia, and East Africa.
- Review First Class and Second Class shipping operations to identify why their late-delivery rates are high.
- Improve delivery-time estimates by comparing scheduled and actual shipping durations to set more realistic delivery commitments.
- Monitor high-volume product categories such as Leather, Men's Footwear, and Women's Apparel to identify fulfillment bottlenecks.
- Prioritize broader logistics improvements since late-delivery rates are consistently high across Consumer, Corporate, and Home Office segments.
- Monitor pending and incomplete orders to identify payment and order-processing bottlenecks.
- Track delivery KPIs regularly to measure improvements in on-time delivery, late-delivery rates, and actual shipping time.

DASHBOARD OVERVIEW



CONCLUSION

The analysis of the DataCo Supply Chain dataset reveals that delivery delays are the major operational challenge, with late deliveries accounting for more than half of the orders. The problem is not limited to a single region, product category, or customer segment, indicating that broader improvements in logistics and fulfillment operations are required.

Delivery performance also varies across shipping modes and regions. First Class and Second Class shipments show particularly high late-delivery rates, while Same Day shipping performs comparatively better. The comparison between scheduled and actual shipping days further indicates inconsistencies in delivery-time estimation and execution.

The findings suggest that the company should focus on improving logistics in high-risk regions, reviewing shipping-mode performance, increasing the accuracy of delivery estimates, and monitoring high-volume product categories. Regular tracking of delivery, order-status, and fulfillment KPIs can help identify bottlenecks early, reduce delays, improve customer satisfaction, and increase overall supply chain efficiency.